Data Activity 3: COVID-19 Data Categorisation and Frequency Analysis

```
> # Upload file
> covid19cases <- read.csv(file.choose())
>
> # Data set structure
> str(covid19cases)
'data.frame':     270 obs. of  7 variables:
 $ Sno                    : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Date                   : chr  "30-01-2020" "31-01-2020" "01-02-2020"
"02-02-2020" ...
 $ State.UnionTerritory   : chr  "Kerala" "Kerala" "Kerala" "Kerala" ...
 $ ConfirmedIndianNational : int  1 1 2 3 3 3 3 3 3 3 ...
 $ ConfirmedForeignNational: int  0 0 0 0 0 0 0 0 0 0 ...
 $ Cured                  : int  0 0 0 0 0 0 0 0 0 0 ...
 $ Deaths                 : int  0 0 0 0 0 0 0 0 0 0 ...
>
>
> # Corrects date format as it initially appears as a character format
> covid19cases$Date <- as.Date(covid19cases$Date, format = "%d-%m-%Y")
>
>
> # Task 1 Create a binary variable called has_deaths that indicates whether any
deaths were reported:
> # - Value = 1 (or TRUE) if Deaths > 0
> # - Value = 0 (or FALSE) if Deaths = 0
>
> covid19cases$has_deaths<- ifelse(covid19cases$Deaths> 0, 1, 0)
>
>
> # Task 2 Create a frequency table using the table() command to analyze death
reporting patterns.
> table(covid19cases$has_deaths)

  0   1
245  25
>
>
> # Task 3 Convert to a factor variable with appropriate labels ("No Deaths",
"Deaths Reported").
> covid19cases$has_deaths_factor <- factor(
+   covid19cases$has_deaths,
+   levels = c(0, 1),
+   labels = c("No Deaths", "Deaths Reported")
+ )
> table(covid19cases$has_deaths_factor)

      No Deaths Deaths Reported
            245             25
>
> # Task 4 Create a categorical variable called case_level based on total
confirmed cases (Indian + Foreign nationals):
> # - No Cases = 0 cases
> # - Low Cases = 1-5 cases
> # - Medium Cases = 6-15 cases
> # - High Cases = 16+ cases
>
> # Total Confirmed Cases (Indian National + Foreign Nationals)
> covid19cases$TotalConfirmed <-
+   covid19cases$ConfirmedForeignNational +
+   covid19cases$ConfirmedIndianNational
>
> # Categorical Variable
> covid19cases$case_level <- cut(
+   covid19cases$TotalConfirmed,
```

```
+   breaks = c(-Inf, 0, 5, 15, Inf),
+   labels = c("No Cases", "Low Cases", "Medium Cases", "High Cases"),
+   right = TRUE
+ )
> # Display case level
> table(covid19cases$case_level)

    No Cases    Low Cases Medium Cases   High Cases
           0          177           61           32
>
> # Task 5 Create a frequency table for the State/Union Territory variable to
see which states had the most frequent reporting.
> table(covid19cases$State.UnionTerritory)

                     Andhra Pradesh                              Chandigarh
Chattisgarh                          Chhattisgarh
Delhi                               Gujarat                                  Haryana
Himachal Pradesh
                                   10                                        1
1                                     2                                      20
2                                    18                                       1
                  Jammu and Kashmir                               Karnataka
Kerala                                Ladakh                          Madhya Pradesh
Maharashtra                                Odisha
Pondicherry
                                    1                                       13
52                                     1                                     1
13                                     6                                     2
                          Puducherry                                  Punjab
Rajasthan                           Tamil Nadu
Telengana        Union Territory of Chandigarh Union Territory of Jammu and
Kashmir            Union Territory of Ladakh
                                    2                                       13
19                                    15                                    20
2                                    12                                     14
                       Uttar Pradesh                           Uttarakhand
West Bengal
                                   18                                        7
4
>
> # Task 6 Identify the top 10 states with the highest number of daily reports
in the dataset.
> sort(table(covid19cases$State.UnionTerritory), decreasing = TRUE)

                               Kerala                                   Delhi
Telengana                             Rajasthan
Haryana                            Uttar Pradesh                          Tamil
Nadu            Union Territory of Ladakh
                                   52                                       20
20                                    19                                    18
18                                    15                                    14
                            Karnataka                           Maharashtra
Punjab Union Territory of Jammu and Kashmir                         Andhra Pradesh
Uttarakhand                                 Odisha                           West
Bengal
                                   13                                       13
13                                    12                                    10
7                                      6                                     4
                         Chhattisgarh                                Gujarat
Pondicherry                           Puducherry        Union Territory of
Chandigarh                           Chandigarh
Chattisgarh                       Himachal Pradesh
                                    2                                       2
2                                     2                                     2
1                                     1                                     1
```

```
                  Jammu and Kashmir                                   Ladakh
Madhya Pradesh
                                  1                                        1
1
>
> head(sort(table(covid19cases$State.UnionTerritory), decreasing = TRUE), 10)

                     Kerala                      Delhi                  Telengana
Rajasthan                   Haryana             Uttar Pradesh
Tamil Nadu Union Territory of Ladakh                  Karnataka
Maharashtra
                         52                          20                         20
19                          18                          18                         15
14                          13                          13
>
> # Sort top 10 states into a list
> as.data.frame(
+   head(sort(table(covid19cases$State.UnionTerritory), decreasing = TRUE), 10))
                        Var1 Freq
1                     Kerala   52
2                      Delhi   20
3                  Telengana   20
4                  Rajasthan   19
5                    Haryana   18
6               Uttar Pradesh   18
7                  Tamil Nadu   15
8  Union Territory of Ladakh   14
9                  Karnataka   13
10               Maharashtra   13
```